Week 4 Day 1 — ML Foundations: Summary

Adult Census Income dataset | Problem framing, hold-out test and baselines | Aqsa Saqib

1. Problem framing

- **Goal:** predict if a person earns more than 50K per year. Positive class = >50K.
- **Data:** 32,561 people and 14 features. 7,841 people (24%) earn >50K and 24,720 (76%) earn <=50K.
- **Business goal:** contact high earners for a premium product without wasting money on wrong people.
- **Split:** 80% train (26,048 rows) and 20% hold-out test (6,513 rows), stratified, random_state = 42. The test set is used only for the final check.

2. Chosen metric: F1 score

- **Precision** = how many selected people are really high earners. Low precision wastes money.
- **Recall** = how many high earners we found. Low recall misses customers.
- Both mistakes matter here, so I use **F1**, which balances precision and recall.
- I do not use accuracy, because predicting <=50K for everyone already gives 76% accuracy.

3. Baseline results (hold-out test set)

Baseline	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority class (always <=50K)	0.759	0.000	0.000	0.000	0.500	0.241
Rule: education.num >= 13	0.758	0.498	0.511	0.505	0.674	0.372

Confusion matrix	True negative	False positive	False negative	True positive
Majority class	4,945	0	1,568	0
Rule: education.num >= 13	4,136	809	766	802

The education rule wins. The majority baseline has 76% accuracy but F1 = 0, because it never finds a high earner. The education rule has almost the same accuracy and F1 = 0.51. A real model must score clearly more than 0.51 to be useful.

4. Error analysis (education rule)

- **False positives (809):** have a degree but earn <=50K. They are younger (average age 38), only 32% are married, and most have only a Bachelors degree.
- **False negatives (766):** no degree but earn >50K. They are older (average age 45), 88% are married, and they work more hours (about 45 per week). Common jobs: Craft-repair, Exec-managerial, Sales.
- **Lesson:** education alone is not enough. Age, marital status, hours and occupation also matter.

5. Next steps

- Fill missing values in workclass, occupation and native.country.
- Change text columns to numbers (one-hot encoding).
- Fix the skewed columns capital.gain and capital.loss.
- Drop education (same as education.num) and fnlwgt (not useful).

Primary metric for the week: F1 score. Score to beat = 0.51 (education rule). My target is F1 of 0.60 or more.